

Table 1: Description of the “Market architecture” pillar using the Theoretical Market Framework for case example

Feature	Sub-feature	Market 1: LT-D-P-A	Market 2: LT-D-P-AE	Market 3: ST-D-P-E	Market 4: RT-D-P-E	Additional Information
Submarkets	Number of submarkets	4				
	Gate Opening Time (GOT)	From months to weeks ahead	From months to weeks ahead	Day-ahead or the next hour after the DSO request (limit 11 pm)	The day of delivery Next hour market for intraday service	
	Gate Closure Time (GCT)	2 days before activation time	2 days before activation time	Day-ahead Day ahead market closes at 14:00 for the day ahead product or next hour market for intraday service	Near Real time	
	Market Time Unit (MTU)	1 hour	1 hour	1 hour	1 hour	quarter of hour is under consideration
	Sub-market type	Auction market	Auction market	Auction market	Auction market	
Services	Service	Congestion management	Congestion management	Congestion management	Congestion management	
Product	Type of product	Active power Availability	Active power Availability Active power Availability and activation	Active Power Activation Active power Availability (optional)	Active Power Activation	
	Technical requirements	As defined for Predictive long-term local active	As defined for Predictive long-term local active	As defined for Predictive short-term local active	As defined for Corrective local active	
Location	Level of spatial granularity	Distribution system areas	Distribution system areas	Distribution system areas	Distribution system areas	Two kinds of areas are considered: Basic areas are a single postal code. Aggregated areas combine basic areas according to DSOs’ needs.
	Responsible System Operator	DSO	DSO	DSO	DSO	
	Voltage Level where resources are located	MV, LV	MV, LV	MV, LV	MV, LV	
Market Roles and actors	Who is the buyer(s)	DSO	DSO	DSO	DSO	
	Who is the seller(s)	SP unit and Group	SP unit and Group	SP unit and Group	SP unit and Group	
	Allowed technologies (Generators, Loads, Storage)	Generators, Loads, Storage	Generators, Loads, Storage	Generators, Loads, Storage	Generators, Loads, Storage	
	Aggregation method	Area	Area	Area	Area	
	Aggregation mix allowed	All technologies, but upward and downward flexibility cannot be aggregated in the same bid	All technologies, but upward and downward flexibility cannot be aggregated in the same bid	All technologies, but upward and downward flexibility cannot be aggregated in the same bid	All technologies, but upward and downward flexibility cannot be aggregated in the same bid	
	Who is the MO	IMO	IMO	IMO	IMO	
	Participation in submarket	Optional	Optional	Hybrid	Optional	

Table2: Description of the 'Sub-market coordination' pillar of the Theoretical Market Framework of the Short-Term markets designed for case example

Feature	Sub-feature	Market 1: LT-D-P-A	Market 2: LT-D-P-AE	Market 3: ST-D-P-E	Market 4: RT-D-P-E	Additional Information
Allocation principle of flexibility	System operators order	Exclusivity for DSO	Exclusivity for DSO	Exclusivity for DSO	Exclusivity for DSO	
	TSO access to DERs	Not applicable	Not applicable	Not applicable	Not applicable	
	Commitment to bid selection	Not Applicable	Not Applicable	From LT-D-P-A: Conditional From LT-D-P-AE: Conditional	Not Applicable	Participation forwarding based on the condition established in the availability bid
	Forwarding of bids	Not Applicable	Not Applicable	Not Applicable	Not Applicable	Bid forwarding from ST-D-P-E to intraday energy auction has been conceptually investigated
Timeframe for coordination	Market phase for coordination between submarkets	With ST-D-P-E: Procurement	With ST-D-P-E: Procurement	With LT-D-P-A: Procurement With LT-D-P-AE: Procurement	Not Applicable	

Table3: Description of the 'Market optimisation' pillar of the Theoretical Market Framework of the Short-Term markets designed for case example

Feature	Market 1: LT-D-P-A	Market 2: LT-D-P-AE	Market 3: ST-D-P-E	Market 4: RT-D-P-E	Additional Information
Market optimisation	Decentralised	Decentralised	Decentralised	Decentralised	
Submarkets optimisation strategy	Sequential	Sequential	Sequential	Sequential	
Sub-market clearing objective	Minimisation of cost	Minimisation of cost	Minimisation of cost	Minimisation of cost	

Table 4: Description of the 'Market operation' pillar of the Theoretical Market Framework of the Short-Term markets designed for case example

Feature	Sub-feature	Market 1: LT-D-P-A	Market 2: LT-D-P-AE	Market 3: ST-D-P-E	Market 4: RT-D-P-E	Additional Information
Remuneration scheme		Auctions pay-as-bid.	Auctions pay-as-bid.	Auctions pay-as-bid.	Auctions pay-as-bid.	
Remunerated product attribute		Active power Availability	Active power Availability Active power availability and activation	Active Power Availability Active Power Activation	Active Power Activation	
Market clearing type		Discrete	Discrete	Discrete	Discrete	
Procurement frequency		Event-based on DSO call	Event-based on DSO call	Event-based on DSO call	Event-based on DSO call	
Bid properties	Minimum bid size	10 kW	10 kW	10 kW	10 kW	
	Bid structure	Simple	Simple	Simple	Simple	

Table 5: Description of the 'Network representation' pillar of the Theoretical Market Framework of the Short-Term markets designed for case example

Feature	Market 1: LT-D-P-A	Market 2: LT-D-P-AE	Market 3: ST-D-P-E	Market 4: RT-D-P-E	Additional Information
Representation of network constraints	Comprehensive grid data; Partial grid data	Comprehensive grid data; Partial grid data	Comprehensive grid data; Partial grid data	Comprehensive grid data; Partial grid data	
Timing of the inclusion of network constraints	Definition of procurement areas Procurement phase	Definition of procurement areas Procurement phase	Definition of procurement areas Procurement phase	Definition of procurement areas Procurement phase	

Case Example market architecture mapped out using structured tables based on theoretical market framework

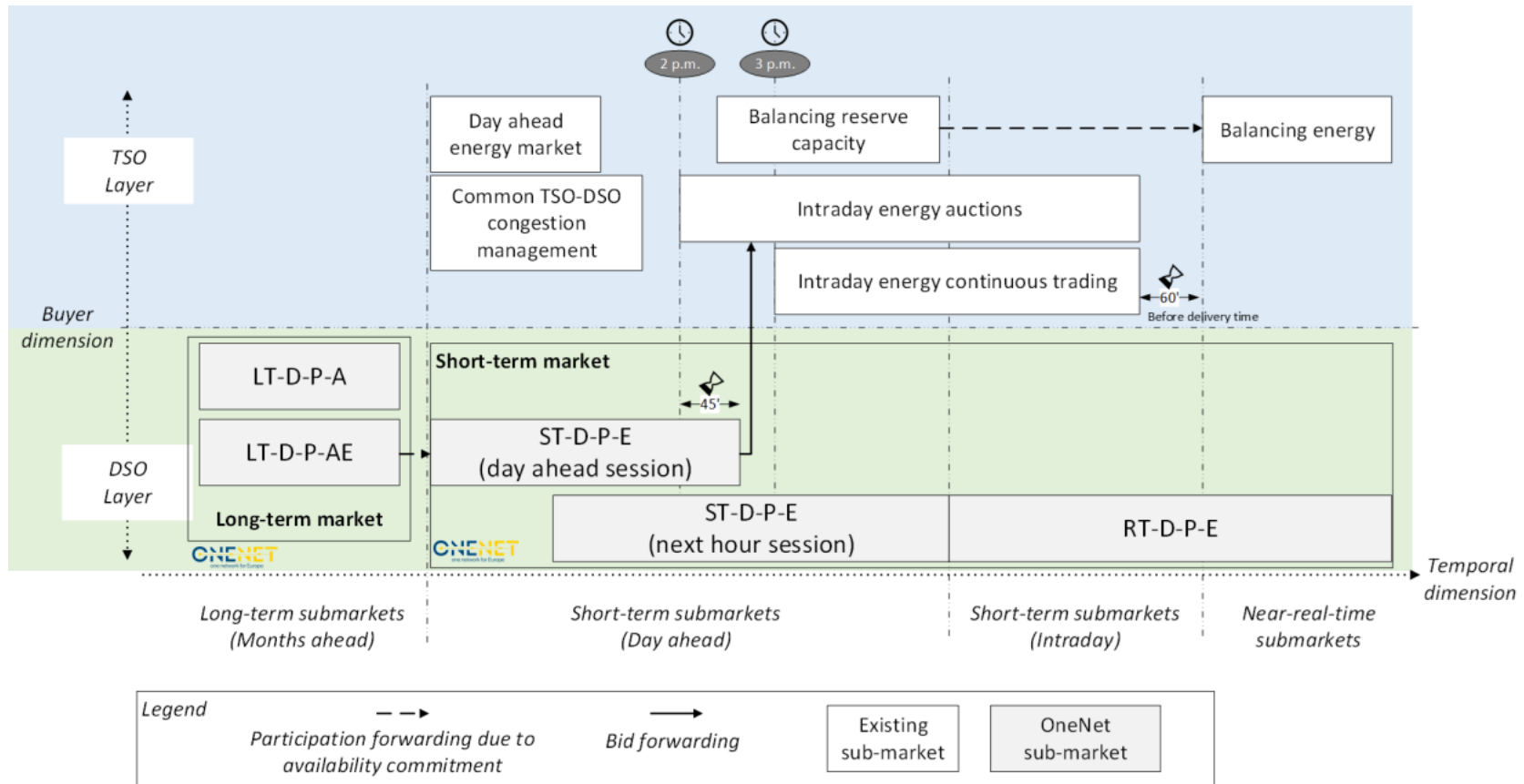


Figure 1: Overview of the Market Architecture for the case example